Bump'em tower: printed parts

Profile: 90x90 light, B-type (Bosch-compatible), slot 10, 2 slots per face 45 mm apart

Fasteners: M6 ISO 7380 button head (4 mm hex) + existing slot-10 T-nuts; plates 5 mm thick

Material: PETG or ASA (not PLA: softens near warm drivers / supplies). No supports needed.

Device sizes: Futek IAA100 drawing FI1573, Mean Well RS-15 case 971A (docs/reference/).

48 V supply RSP-2000-48 (295x127x41 mm, 1.95 kg, case 952D): 2 printed end brackets, 10 mm air gap, ASA

Source: BumpemServer/hardware/enclosures (CadQuery). Dimensions in tower.py.

Part	Status	Still needed
Slot test plates	**READY TO PRINT**	nothing
Base mount plate	**READY (geometry)**	test-fit result (KEY_CLEAR), bolt length
IAA100 DIN-rail mount	**READY (datasheet)**	check your IAA100 has the DIN clip (66.8 x 50.4 x 17.8 mm version)
24 V supply plate (RS-15-24)	**READY (datasheet)**	M3 x 6 screws; a cover for the open 230 V terminals (electrician)
48 V supply brackets (RSP-2000-48)	**READY (datasheet)**	M4 x 12 screws; check the supply's 3 M4 holes on the wide face match (case 952D)
Signal connector plate	**CONCEPT**	MSTB panel part + cut-out size
Cable clip	**CONCEPT**	cable diameters (PLACEHOLDER 8 mm)
Power-strip hook (x2)	**CONCEPT**	power strip L x W x H

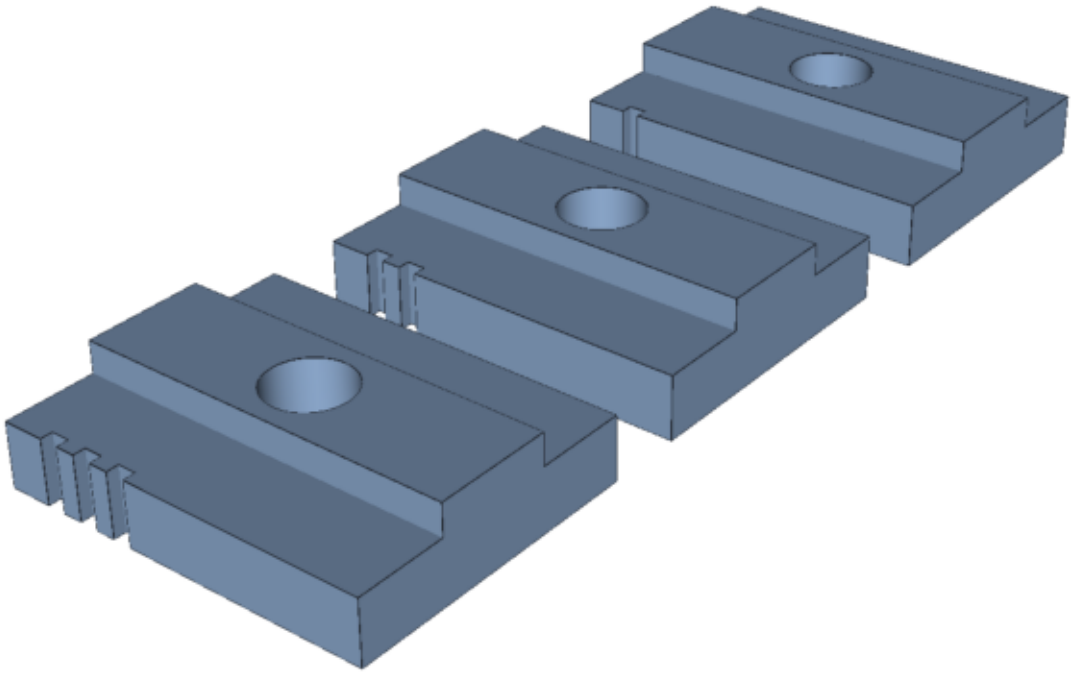
Order: 1) print the slot test plates, 2) report which fits, 3) measure the devices marked PLACEHOLDER, 4) final holders are generated from those numbers.

Mains: the power strip is only carried by printed hooks; no 230 V parts inside printed housings.

Slot test plates

READY TO PRINT

Part (as modelled; print orientation in the STL)



What: 3 small plates, key clearance 0.1 / 0.2 / 0.3 mm (1/2/3 notches). Sets KEY_CLEAR for all parts.

Size: 30 x 84 x 8 mm

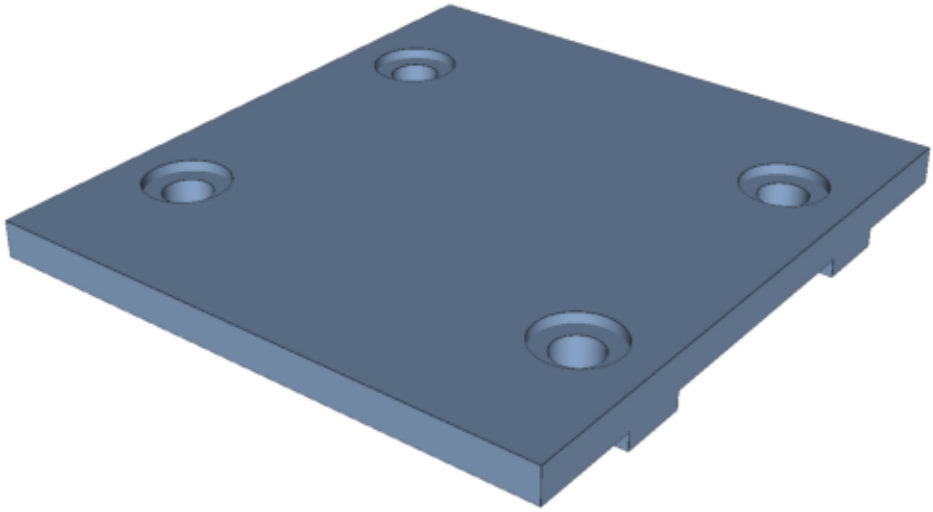
Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

Still needed: nothing

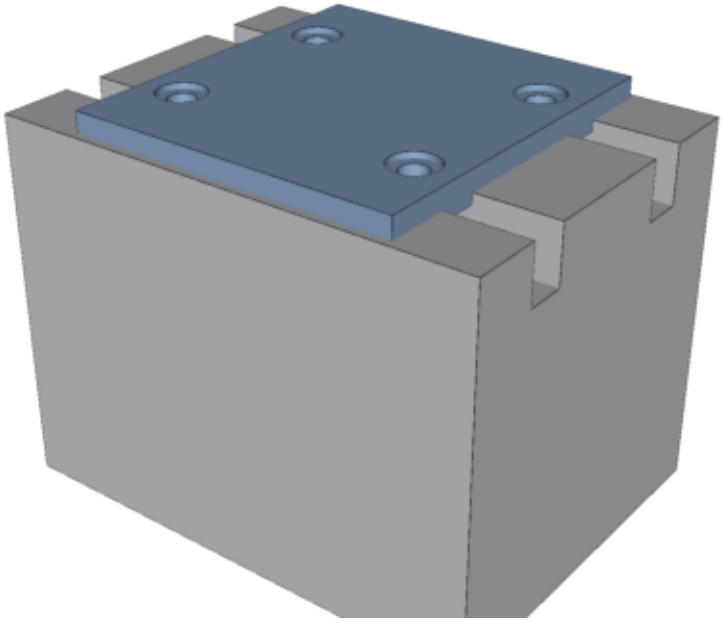
Base mount plate

READY (geometry)

Part (as modelled; print orientation in the STL)



Mounted on the profile (grey); device envelope in red



What: Covers one face, 2 keys in both slots, 4 M6 holes with head seats. Template for wider parts.

Size: 80 x 86 x 8 mm

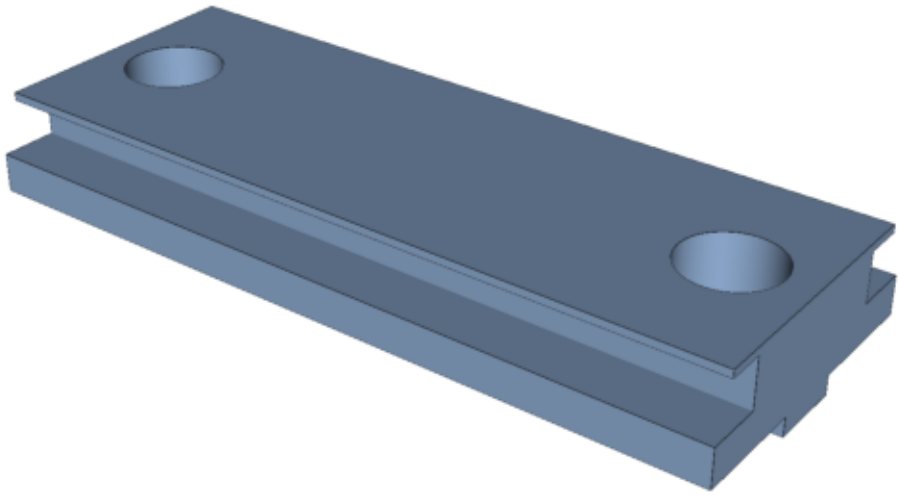
Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

Still needed: test-fit result (KEY_CLEAR), bolt length

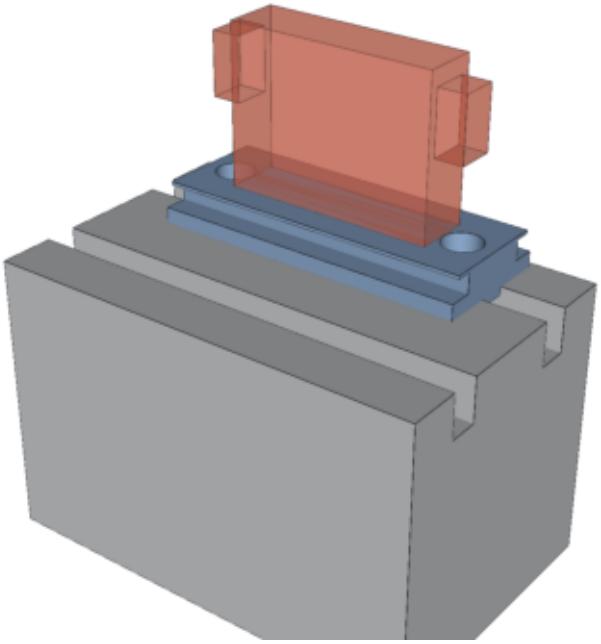
IAA100 DIN-rail mount

READY (datasheet)

Part (as modelled; print orientation in the STL)



Mounted on the profile (grey); device envelope in red



What: 100 mm printed 35 mm top-hat rail on a one-slot plate. The Futek IAA100 clips on (spring DIN clip, FI1573); plugs point up and down along the profile. Alternative: bolt a short steel DIN rail instead.

Size: 100 x 40 x 15 mm

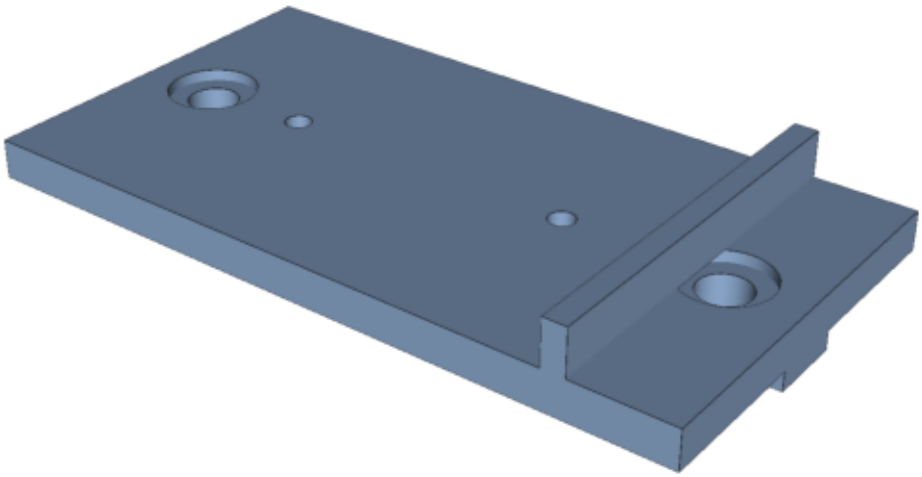
Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

Still needed: check your IAA100 has the DIN clip (66.8 x 50.4 x 17.8 mm version)

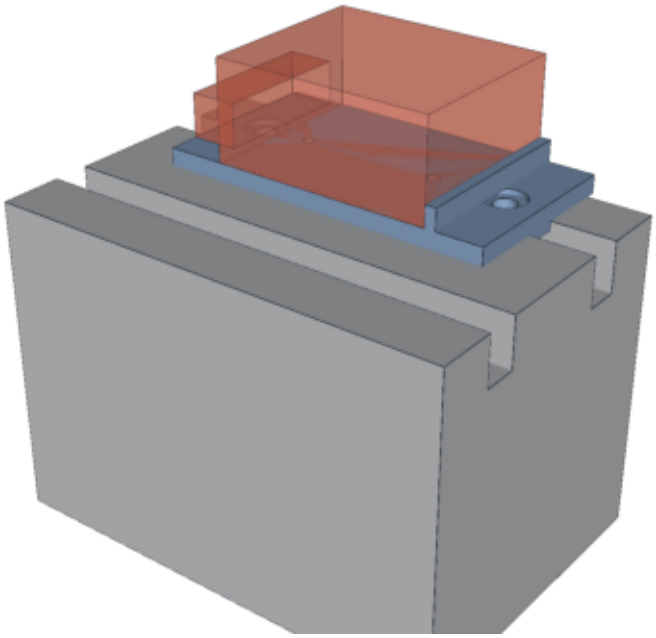
24 V supply plate (RS-15-24)

READY (datasheet)

Part (as modelled; print orientation in the STL)



Mounted on the profile (grey); device envelope in red



What: Plate under the Mean Well RS-15-24 (62.5 x 51 x 28 mm, case 971A), held by its 2 bottom M3 holes with M3 x 6 screws from the back; end stop opposite the terminals.

Size: 95 x 51 x 14 mm

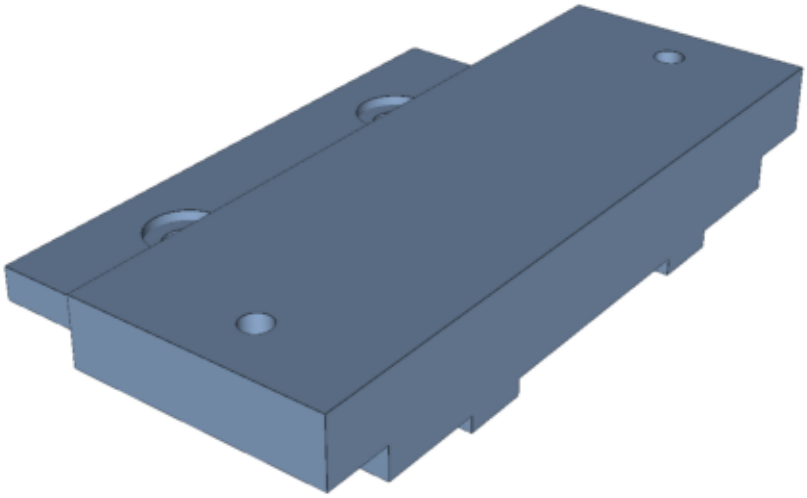
Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

Still needed: M3 x 6 screws; a cover for the open 230 V terminals (electrician)

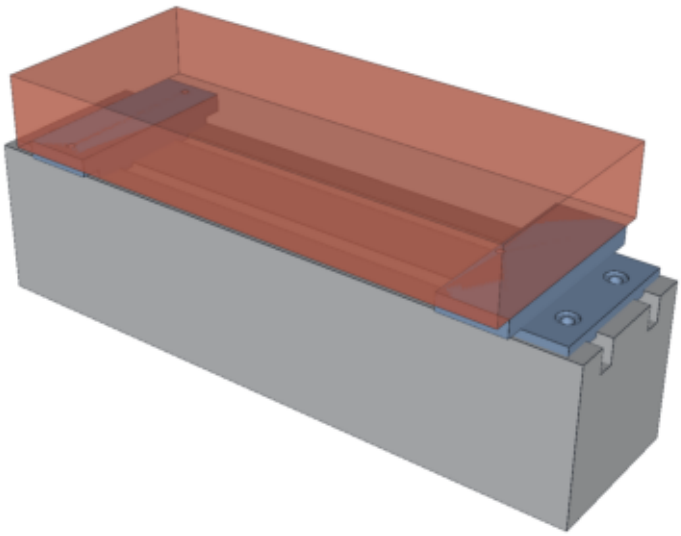
48 V supply brackets (RSP-2000-48)

READY (datasheet)

Part (as modelled; print orientation in the STL)



Mounted on the profile (grey); device envelope in red



What: Two end brackets (A: 2 x M4, B: 1 x M4) under the Mean Well RSP-2000-48 (295 x 127 x 41 mm, 1.95 kg, case 952D). Keyed into both slots; M6 bolts past the supply ends stay reachable; 10 mm air gap behind the supply; fan ends free. Screw the supply onto both brackets first (M4 x 12 from the back, max 4 mm into the case), then bolt them to the profile. Print in ASA or PETG.

Size: 62 x 107 x 18 mm

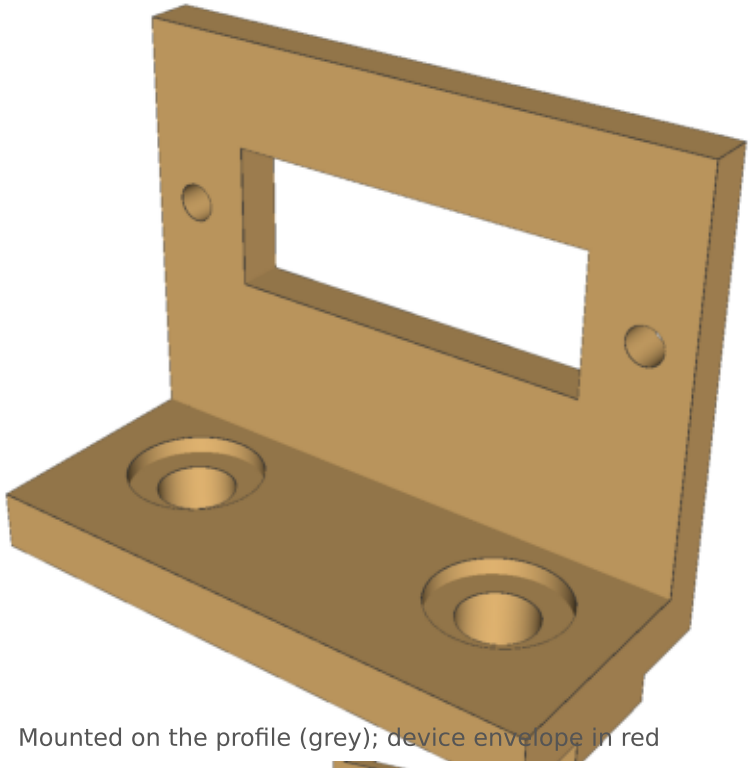
Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

Still needed: M4 x 12 screws; check the supply's 3 M4 holes on the wide face match (case 952D)

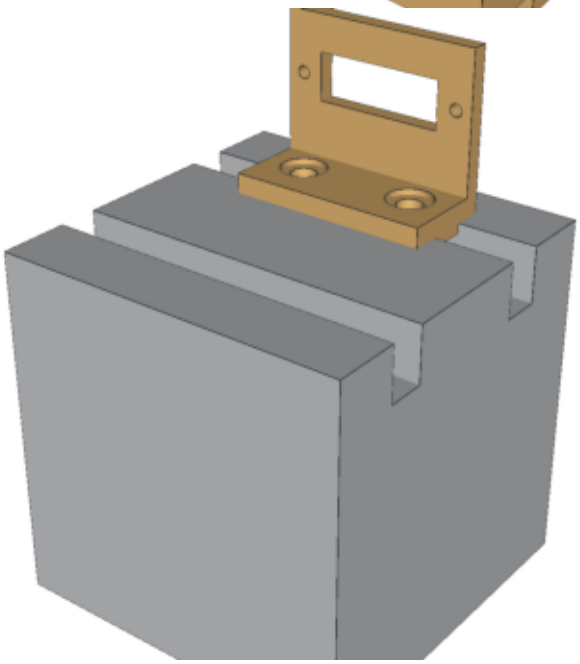
Signal connector plate

CONCEPT

Part (as modelled; print orientation in the STL)



Mounted on the profile (grey); device envelope in red



What: Upright panel holding the tower's 5-pin MSTB socket: the single signal cable to the breadboard / carrier board.

Size: 50 x 24 x 43 mm

Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

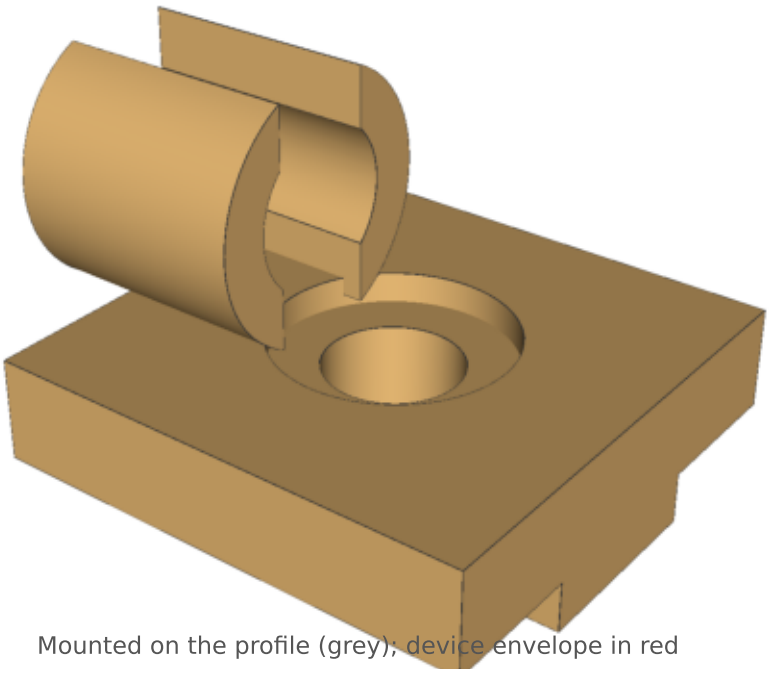
Still needed: MSTB panel part + cut-out size

Concept only: device sizes are placeholders, the final part will change.

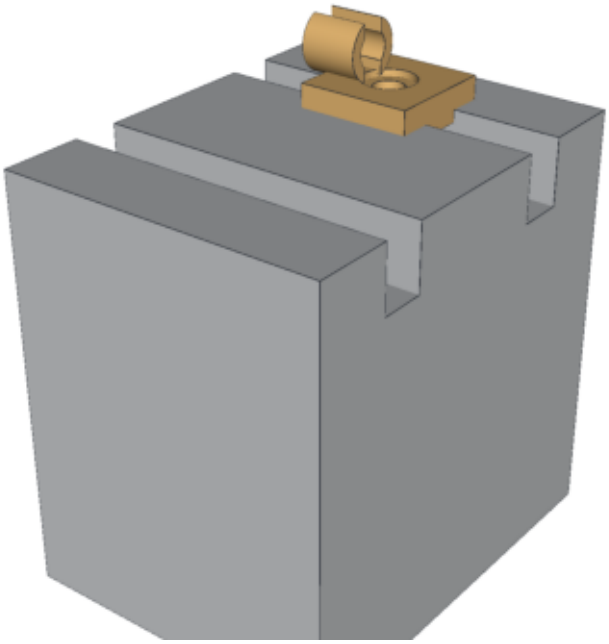
Cable clip

CONCEPT

Part (as modelled; print orientation in the STL)



Mounted on the profile (grey); device envelope in red



What: Snap-in clip guiding a cable along the profile (mains, signal, motor).

Size: 28 x 24 x 18 mm

Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

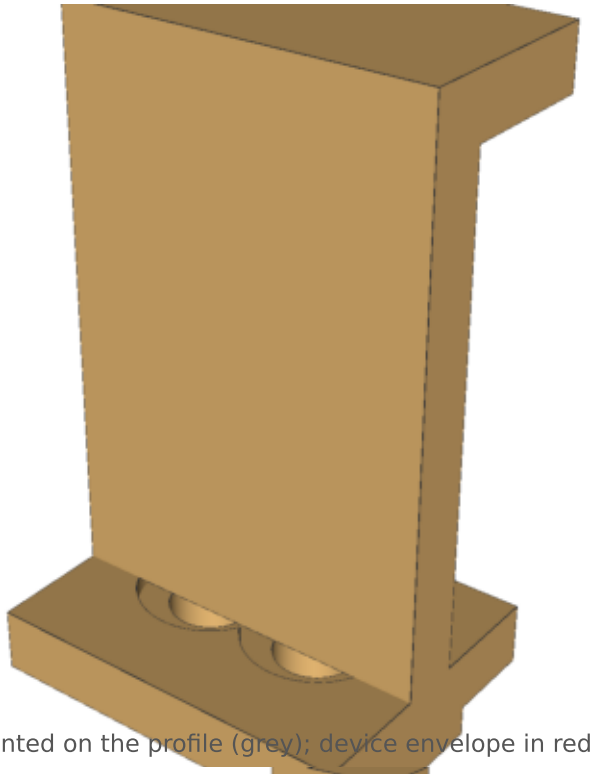
Still needed: cable diameters (PLACEHOLDER 8 mm)

Concept only: device sizes are placeholders, the final part will change.

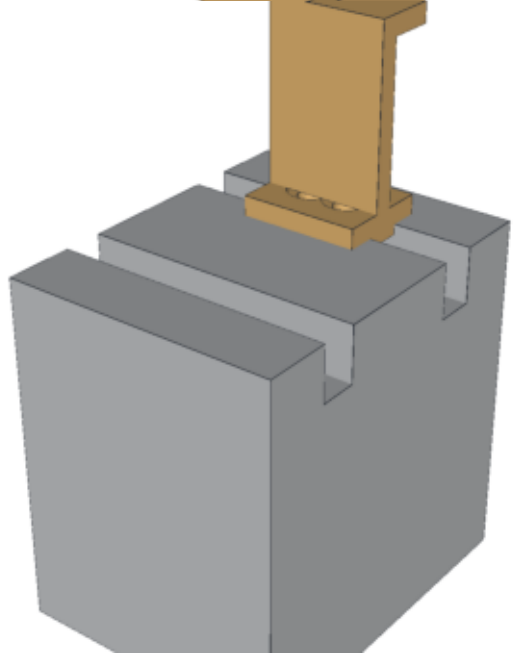
Power-strip hook (x2)

CONCEPT

Part (as modelled; print orientation in the STL)



Mounted on the profile (grey); device envelope in red



What: Two hooks carrying the tower's power strip. Holder only, no mains parts inside the print.

Size: 30 x 26 x 53 mm

Mount: M6 button-head screws in slot-10 T-nuts; key 9.6 mm in the slot

Still needed: power strip L x W x H

Concept only: device sizes are placeholders, the final part will change.